Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Status Update 5

Team 12: TI-Amplifier
Braeden Murdock
Cole Keating
Haotian Guan
Jonathan Roman
TA: Manoj Kumar Balamurugan
Sponsor: Texas InstrumentS

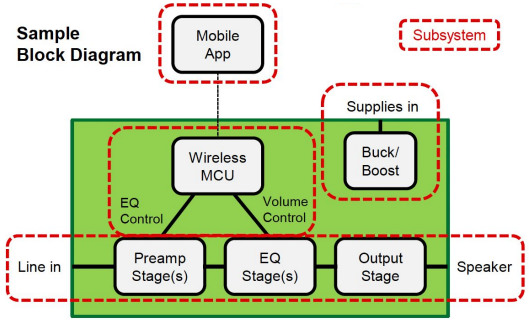
Project Summary

Problem statement:

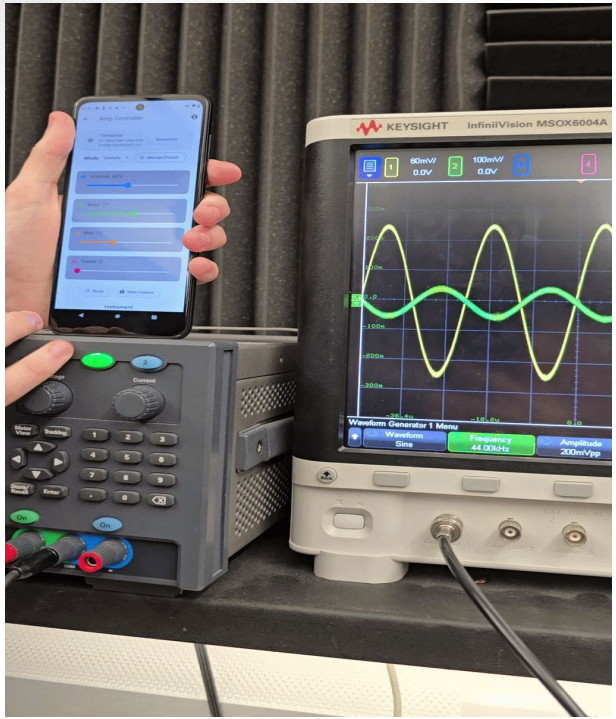
- Needs a system that can wirelessly amplify the sound of a 8Ω speaker from a mobile app.

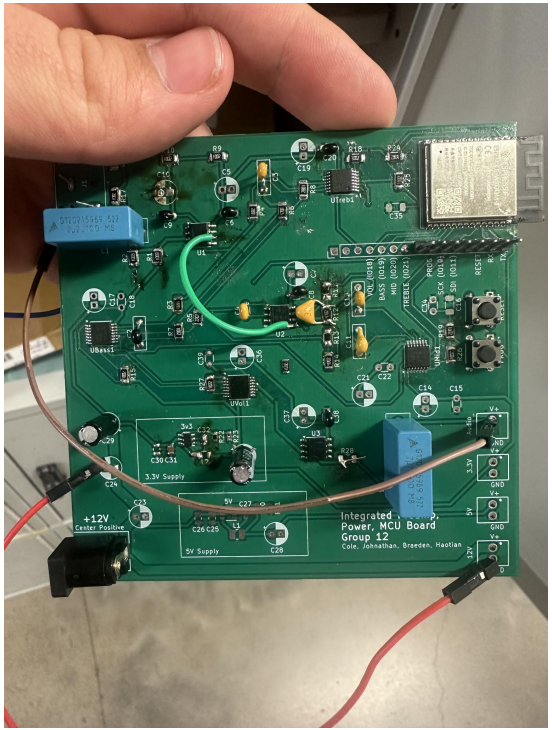
Our Solution:

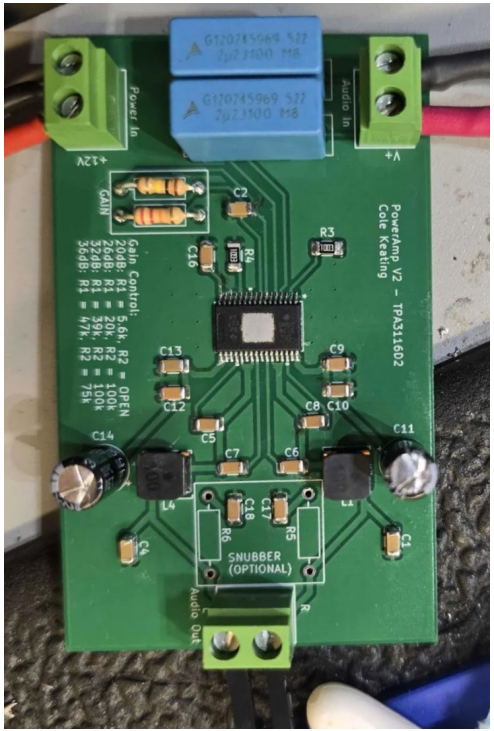
- The system being developed will use a MCU to communicate with a mobile app in order to control a system of amplifiers to adjust volume and EQ of the speaker using a aux input.



Integrated System Diagram







Project Timeline

Subsystem Designs and Testing (completed 1/26)	Integration of Mobile App and MCU (completed 2/2)	Complete first revision of integrated PCB (completed 2/16)	System Test (completed 3/2)	Completion of Final Integrated PCB (completed 3/16)	Validation (to complete by 4/6)	Demo and Report (to complete by 4/13)

Subsystem 1 - Mobile App

Haotian Guan

Accomplishments since last presentation 17 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed an background theme selection UI • Fixed the issue where control page keep returning back to connection page in skip connection mode. • Builded an apk.	• Continue testing and debugging

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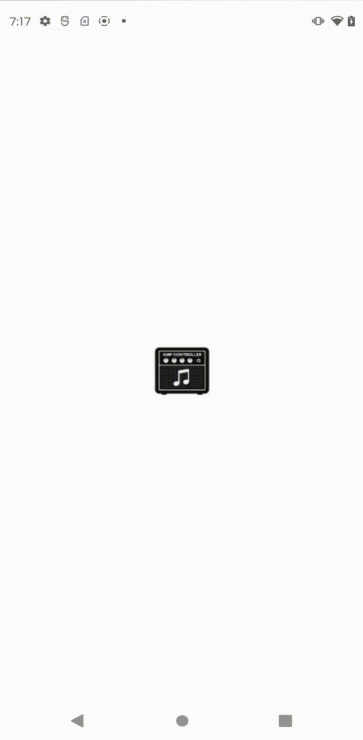
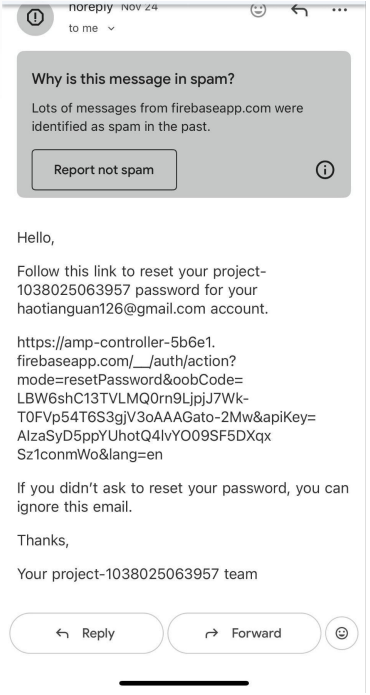
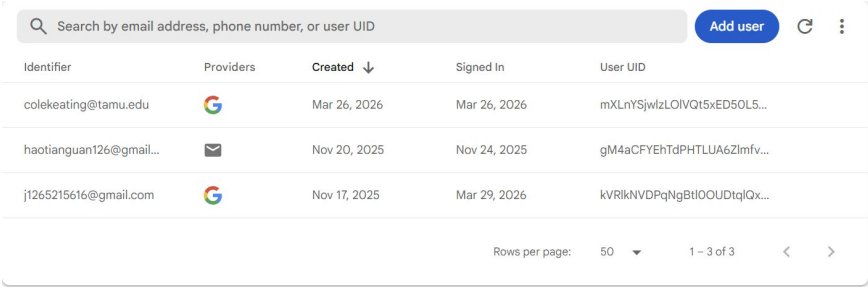
Subsystem 1 - Mobile App

Haotian Guan







Subsystem 1 - Mobile App

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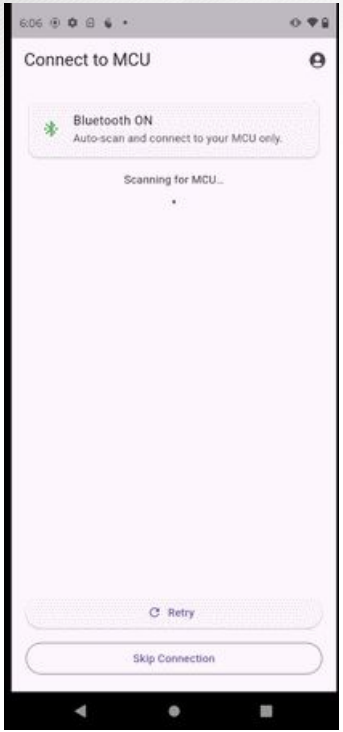


```
D/[FBP-Android](17452): [FBP]   status: 0GATT_SUCCESS
D/BluetoothGatt(17452): onConnectionUpdated() - Device=74:4D:BD:68:61:3E interval=39 latency=0 timeout=500 status=0
D/[FBP-Android](17452): [FBP] onMethodCall: setNotifyValue
D/BluetoothGatt(17452): setCharacteristicNotification() - uuid: 00002a05-0000-1000-8000-00805f9b34fb enable: true
D/[FBP-Android](17452): [FBP] onDescriptorWrite:
D/[FBP-Android](17452): [FBP]   chr: 2a05
D/[FBP-Android](17452): [FBP]   desc: 2902
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
I/flutter (17452): [BLE WRITE] port=2 value=13
I/flutter (17452): [FBP] stopScan: already stopped
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
I/flutter (17452): [BLE WRITE] port=0 value=30
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
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I/flutter (17452): [BLE WRITE] port=1 value=0
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
I/flutter (17452): [BLE WRITE] port=2 value=0
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
I/flutter (17452): [BLE WRITE] port=3 value=0
D/BluetoothGatt(17452): onClientConnectionState() - status=0 clientIf=6 device=74:4D:BD:68:61:3E
D/[FBP-Android](17452): [FBP] onConnectionStateChange:disconnected
D/[FBP-Android](17452): [FBP]   status: SUCCESS
D/BluetoothGatt(17452): close()
```

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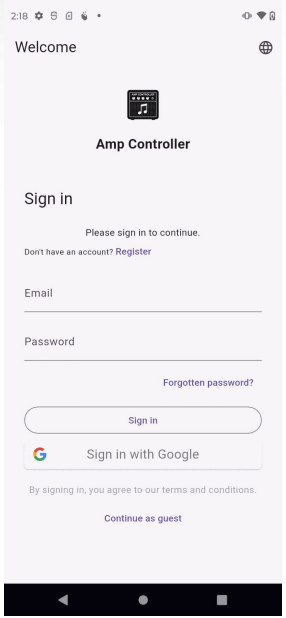
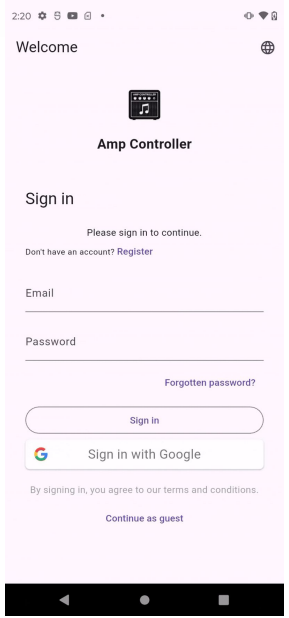
Subsystem 1 - Mobile App

Haotian Guan

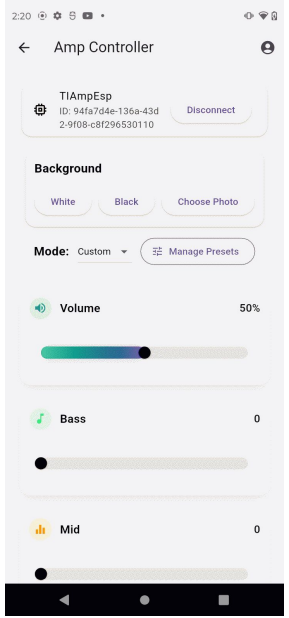
Localization



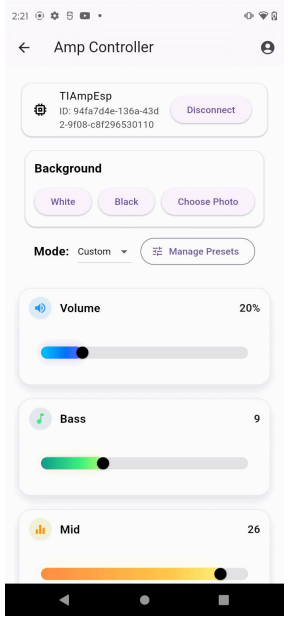
Volume and EQ Control



Preset and Mode



Theme and Piano



Subsystem 2 - Microcontroller

Jonathan Roman

Accomplishments last presentation 20 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Validated MCU operations on integrated PCB • Assisted in testing for other integrated subsystems	• Continue Assisting in Pre-amp validation • Refine and Debug code for new ESP model • Design housing for full system using CAD

Subsystem 3 - Microcontroller

Jonathan Roman

Previous Iteration of PCB was used to find all bugs for New PCB the list of bugs found and solutions are as follow

Bug	Solution
Incompatible header files for current integration	Replaced relevant header files and updated code
Connection Errors leading to bad signals	Resolder key connections and verify functionality
BLE Connection Issues	Formatting differences overlooked in header transfer

```
#include <BLEDevice.h>
#include <BLEUtils.h>
#include <BLEServer.h>
```

```
#include <NimBLEDevice.h>
```

Subsystem 3 - Power System

Braeden Murdock

Accomplishments last presentation 26 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Validated Power System on new PCB to make sure it worked as expected • Helped in Debugging Current PCB noise Issue with Cole	• Continue Assisting in Pre-amp validation • Help Jonathan in building the housing for entire system using CAD

Subsystem 3 -Power System

Braeden Murdock

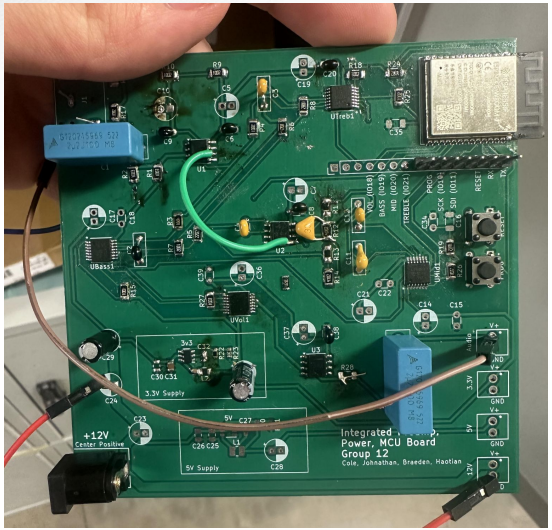
Previous Iteration of PCB was used to find all bugs for New PCB the list of bugs found and solutions are as follow

Bug	Solution
MCU failed to send signal to digital pots	Digital pots required 16 total bits sent across two 8 bit signals instead of the 1 8 bit we were sending before
Digital Pot held high at 5 V when should be at 3.3V	MCU pin was burned out due to prior testing and caused it to send a constant read command instead of write command

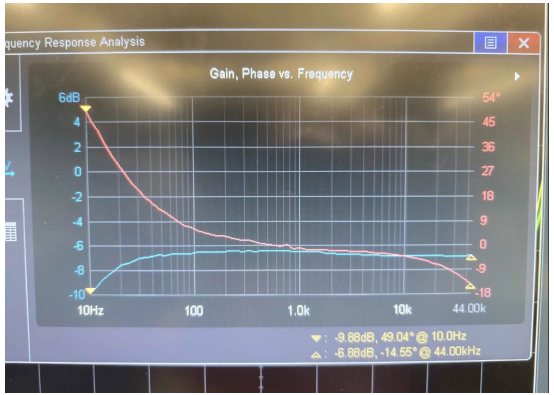
Expected Power on Rail	Actual Power on Rail
12V	12.43V
3.3V	3.371V

Subsystem 4 - Analog Section

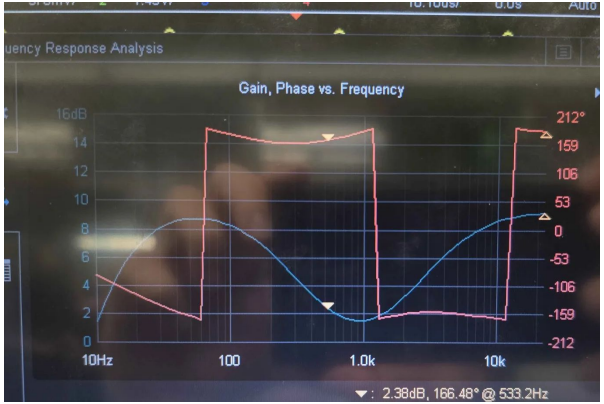
Cole Keating



Newest PCB revision with HV pots



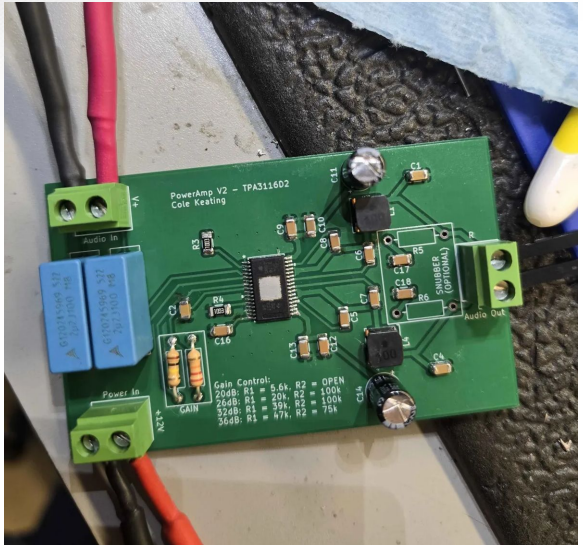
Flat response through pots - no passive degradation of audio signal



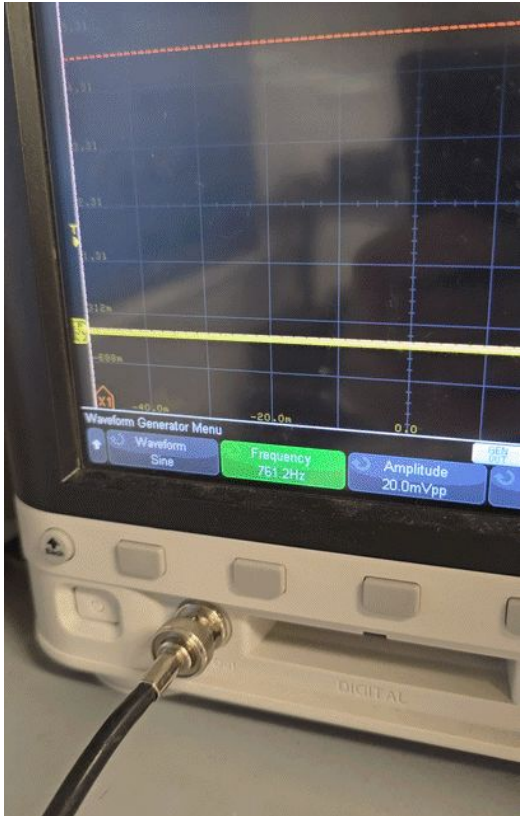
Testing of increased gain in bass, treble frequencies

Subsystem 4 - Power Amp

Cole Keating



Finished Power Amp



Subsystem 4 - Analog Section

Cole Keating

Accomplishments since last update 20 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Corrected major issue with oscillation in mid stage • Validated signal integrity through preamp section • Validated efficacy of digital pots in audio role • ONGOING - Validate increased gain using high voltage pots	• Validate all past tests on new iteration of PCB - minimal changes, don't anticipate issues • Need to order high wattage load resistor to validate maximum gain of power amp

Execution Plan

Task	Winter Break	01/12/26	01/19/26	01/26/26	02/02/26	02/09/26	02/16/26	02/23/26	03/02/26	03/09/26	03/16/26	03/23/26	03/30/26	04/06/26	04/13/26	04/20/26	Date
MCU PCB Design Completed																	
403 Status Verification																	
Power and Amplifier Integration																	
Initial Bluetooth Connection																	
MCU PCB Assembly																	
Initial Status Presentation																	
Kicad -> Altium Translation																	
Integrated PCB Design																	
Debug and optimize UI																	
Debug Power Amp Design																	
MCU PCB Validation																	
Test Power Amplifier																	
Integrated PCB Assembly																	
Integrated PCB Testing																	
Full System Assembly																	
Casing CAD Design																	
Finalize and Debug Project																	
Engineering Project Showcase																	04/24/26

Not Started
In Progress
Completed
Behind Sch.

Validation Plan

Identifier	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.3.1	Verify input signal levels	Based on research, standard line level signal is 1.4Vpeak, need to ensure this as this is what preamp is biased around	Connect 3.5mm audio cable to source computer, Measure peak voltages to ensure amp isn't being over or underdriven	Pass	Cole Keating
3.2.3.2	Measure power amp heat dissipation	TPA3116 should not enter thermal protection during test	Place cooling solution on chip, power a 8 Ohm load for 30+ minutes without the FAULT pin on the IC going high (indicating thermal protection)	Pass	Cole Keating
3.2.3.3	Verify preamp output signal levels	Preamp output should never exceed 3Vpeak - From MAX9744 datasheet. Signal at max gain should not exceed this	Make sure 0dB bias level is below 10dB of 3V. Set input signal to 1.4V at a mixer frequency. Move potentiometer for max gain, verify output does not exceed 3V peak on oscilloscope.	Pass	Cole Keating
3.2.3.4	Verify power amp output signal level	Power amp output should be able to drive a 8 ohm speaker at 5W without clipping.	Place load on output, verify output levels do not clip.	Pass	Cole Keating
3.2.3.5	Verify THD parameters	THD < 5% at 1WRMS, THD <0.1% at 0.1W RMS	Use FRA mode on oscilloscope while driving a 1W and 0.1W load. Record THD levels.	Pass	Cole Keating
3.2.3.6, 3.2.4.2	Integration of MCU EQ Control	MCU is able to command digital potentiometer. Digital potentiometer controls preamp mixing, volume.	Connect digital potentiometers to pin headers on preamp. Run MCU test code that sweeps potentiometers. Verify digital potentiometer controls preamp, volume	Pass	Cole Keating Jonathan Roman
3.2.3.7	Verify ideal gain setting	Need to find ideal ratio between volume level, current draw, THD, and thermal output. This is a subjective measure, will need to form consensus with team	Power amp has configurable gain set using resistor divider. Need to find ideal combination relating to signal levels coming from volume pot.	Validating	Cole Keating
3.2.3.8	Verify gain/attenuation levels with 12V biasing	Each band should be able to boost/attenuate 10dB.	Set band to maximum/minimum of adjustment range. Use the FRA tool on oscilloscope to verify dBs of boost/attenuation. Repeat for all 6 possible levels. Max/Min of Low, Mid, High. Measure output waveform with signal generator connected. Output should not clip.	Validating, Verified for 5V board	Cole Keating
3.2.2	Verify Power levels	Based on research and simulations nothing in the circuit should exceed 2 Watts of power	Connect Power management subsystem to Power input and measure all outputs to make sure they are within the criteria	Pass	Braeden Murdock
3.2.2.1	Verify input voltage levels	Each subsystem needs varous input voltage supplies which are 12V, 5V, and 3.3V. The power management subsystem should output all desired levels of voltage	Measure output voltages of power management subsytem when powered on, by measureing the connectors that would lead to the other subsystems that need the Voltages described previously	Pass	Braeden Murdock
3.2.2.2	Verify input current	Each subsystem has current draw that needs to be matched for the system to work. These range from 16 mA, 0.5 A and 4mA	Set up load resistors that will have the required voltage draw and measure the current through each to make sure that the power management subsystem can match	Pass	Braeden Murdock

Validation Plan Continued

3.2.4.1	Verify Wireless Connection	MCU is able to connect wirelessly to a mobile device. Mobile Device is used to send signals translated by the MCU	Power the MCU with a 3.3V Source. Connect through a generic Bluetooth integration. Send and recieve signals	Pass	Jonathan Roman
3.2.4.1	Verify Wireless Connection With Mobile App	MCU is able to connect wirelessly to the Mobile App subsystem. Mobile App is used to send signals translated by the MCU	Power the MCU with a 3.3V Source. Connect through integrated Mobile App. Recieve and output signals	Pass	Hoation Guan Jonathan Roman
3.2.4.2	Verify Signal Translation	MCU correctly translates input signals to respective SPI output.	Connect wirelessly to the MCU. Send preset signal pairs and verify output	Pass	Jonathan Roman
3.2.4.2	Verify PCB implementation	MCU behaves as expected on PCB implementation	Connect wirelessly to the PCB integrated MCU. Send preset signal pairs and verify output	Pass	Jonathan Roman
3.2.1.1	Verify the user feedback	Ensure UI give an feedback when use adjust the app setting.	When sliding the control slider, a small prompt appears below indicating the change has been applied. Additionally, a touch effect is displayed when the slider is touched.	Pass	Haotian Guan
3.2.1.2	Verify the login page	Ensure users can log in using their Google account or register by using the email. Confirm the firebase stored the accounts.	Use Firebase for Flutter to set up a login page with Google accounts. Click “Sign in as guest” to use the app immediately.	Pass	Haotian Guan
3.2.1.3	Verify the BLE connection	Ensure standardized commands sent to the MCU via BLE	Using Flutter's flutter_blue_plus, find the Service UUID and Characteristic UUID of the MCU, and then the App sends 2 bytes [portal, value] to MCU, where portal select valume/bass/mid/treble and value is the user-selected level.	Pass	Haotian Guan Jonathan Roman
3.2.1.4	Verify the BLE detector	Ensure the mobile app UI return to the connection page once disconnection with MCU	Returning back to connection page once the MCU disconnection with mobile app, and exceeding the notification will alert the user.	Pass	Haotian Guan
3.2.1.5	Verify the background theme	Ensure the control background can be changed by using background selection botton	Add a UI for background selection in the control page, allowing users to choose between the white and black themes, or to load their own photos as the theme.	Pass	Haotian Guan

Team 12 – TI-Amplifier

Braeden Murdock
Cole Keating
Haotian Guan
Jonathan Roman

Thank you for your attention!
We appreciate your time and interest in our project.
We're happy to take any questions.